

Lecture 05: Multiple linear regression

Phuong L. Vo

International University

September 28, 2026



Outline

- 1 Multiple regression model
- 2 Multiple regression model in matrix form
- 3 Analysis of variance results
- 4 Inferences about regression parameters
- 5 Estimation of mean response and prediction of new observation
- 6 Testing the General Linear Hypothesis
 - Extra sums of squares
 - Testing for a subset of coefficients
- 7 Hidden extrapolation and multicollinearity

Outline

- 1 Multiple regression model
- 2 Multiple regression model in matrix form
- 3 Analysis of variance results
- 4 Inferences about regression parameters
- 5 Estimation of mean response and prediction of new observation
- 6 Testing the General Linear Hypothesis
 - Extra sums of squares
 - Testing for a subset of coefficients
- 7 Hidden extrapolation and multicollinearity

First-order model with two predictor variables

- The linear (first-order) regression model with two predictor variables X_1 and X_2 is

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \epsilon_i \quad (1)$$

where Y_i denotes the response, X_{i1} and X_{i2} are the values of the two predictor variables in the i th trial.

- If $E\{\epsilon_i\} = 0$, the regression function of the model:

$$E\{Y\} = \beta_0 + \beta_1 X_1 + \beta_2 X_2, \quad (2)$$

which is a plane, called **response plane**.

- E.g., expected market sales (in \$10,000) is estimated from point-of-sale expenditures (X_1 , in \$1,000) and TV expenditures (X_2 , in \$1,000) by $E\{Y\} = 10 + 2X_1 + 5X_2$.
 - If point-of-sale expenditures are increased by \$1,000 while TV expenditures are held constant, expected sales increase by \$20,000.
 - if TV expenditures in a locality are increased by \$1,000 and point-of-sale expenditures are held constant, expected sales increase by \$50,000.

General linear regression model

- Linear regression model with $p - 1$ predictor variables $X_1, \dots, X_{p-1}$

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_{p-1} X_{i,p-1} + \epsilon_i, \quad (3)$$

where

- $\beta_0, \dots, \beta_{p-1}$ are parameters
 - $X_{i1}, X_{i2}, \dots, X_{i,p-1}$ are known constants
 - ϵ_i are independent and $N(0, \sigma^2)$
- Let $X_{i0} = 1$, we can rewrite linear regression model as

$$Y_i = \sum_{k=0}^{p-1} \beta_k X_{ik} + \epsilon_i = X_i' \beta, \quad (4)$$

where β is vector of coefficients and X_i is the vector of predictor values corresponding to i -th observation.

Transform from polynomial model to linear regression model

- Polynomial regression models:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \epsilon_i, \quad (5)$$

- It is a special cases of the general linear regression model, in which $X_{i1} = X_i$ and $X_{i2} = X_i^2$.

Outline

- 1 Multiple regression model
- 2 Multiple regression model in matrix form
- 3 Analysis of variance results
- 4 Inferences about regression parameters
- 5 Estimation of mean response and prediction of new observation
- 6 Testing the General Linear Hypothesis
 - Extra sums of squares
 - Testing for a subset of coefficients
- 7 Hidden extrapolation and multicollinearity

General linear regression model in matrix terms

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad (6)$$

where

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} 1 & X_{11} & \dots & X_{1,p-1} \\ 1 & X_{21} & \dots & X_{2,p-1} \\ \vdots & \vdots & & \vdots \\ 1 & X_{n1} & \dots & X_{n,p-1} \end{bmatrix} \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_{p-1} \end{bmatrix} \quad \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

Row i of X corresponds to i th observation.

- General linear regression model is given by:

$$Y = X\beta + \epsilon.$$

Estimation of coefficients

- Loss function:

$$\begin{aligned}SSE &= \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_{i1} - \dots - \beta_{p-1} X_{i,p-1})^2 \\ &= (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})\end{aligned}$$

- Loss function is a quadratic function. The optimal point (estimator) satisfies:

$$\mathbf{X}'\mathbf{X}\mathbf{b} = \mathbf{X}'\mathbf{Y}$$

Hence, the least squares estimators are:

$$\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

Fitted values and residuals

- The fitted values are

$$\hat{\mathbf{Y}} = \mathbf{X}\mathbf{b} = \mathbf{H}\mathbf{Y}$$

where hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$

- Residual terms are:

$$\begin{aligned}\mathbf{e} &= \mathbf{Y} - \hat{\mathbf{Y}} = \mathbf{Y} - \mathbf{X}\mathbf{b} \\ &= (\mathbf{I} - \mathbf{H})\mathbf{Y}\end{aligned}$$

- The variance-covariance matrix of the residuals is:

$$\text{var}\{\mathbf{e}\} = \sigma^2(\mathbf{I} - \mathbf{H}),$$

which is estimated by

$$s^2\{\mathbf{e}\} = \text{MSE} * (\mathbf{I} - \mathbf{H}).$$

Outline

- 1 Multiple regression model
- 2 Multiple regression model in matrix form
- 3 Analysis of variance results**
- 4 Inferences about regression parameters
- 5 Estimation of mean response and prediction of new observation
- 6 Testing the General Linear Hypothesis
 - Extra sums of squares
 - Testing for a subset of coefficients
- 7 Hidden extrapolation and multicollinearity

Sums of squares and mean squares

- The sums of squares for the analysis of variance in matrix terms are:

$$SSTO = \mathbf{Y}'\mathbf{Y} - \left(\frac{1}{n}\right) \mathbf{Y}'\mathbf{J}\mathbf{Y} = \mathbf{Y}'\left[\mathbf{I} - \left(\frac{1}{n}\right)\mathbf{J}\right]\mathbf{Y}$$

$$SSE = \mathbf{e}'\mathbf{e} = (\mathbf{Y} - \mathbf{X}\mathbf{b})'(\mathbf{Y} - \mathbf{X}\mathbf{b}) = \mathbf{Y}'(\mathbf{I} - \mathbf{H})\mathbf{Y}$$

$$SSR = \mathbf{b}'\mathbf{X}'\mathbf{Y} - \left(\frac{1}{n}\right) \mathbf{Y}'\mathbf{J}\mathbf{Y} = \mathbf{Y}'\left[\mathbf{H} - \left(\frac{1}{n}\right)\mathbf{J}\right]\mathbf{Y},$$

where $\mathbf{J}$ is an $n \times n$ matrix of 1's and $\mathbf{H}$ is the hat matrix.

- $\mathbf{I} - \left(\frac{1}{n}\right)\mathbf{J}$, $\mathbf{I} - \mathbf{H}$, and $\mathbf{H} - \left(\frac{1}{n}\right)\mathbf{J}$ are $n \times n$ symmetric matrices.
- SSTO has $n - 1$ degrees of freedom, SSE has $n - p$ degrees of freedom, and SSR has $p - 1$ degrees of freedom.

ANOVA table for general linear regression model

Source of Variation	SS	df	MS
Regression	$SSR = \mathbf{b}'\mathbf{X}'\mathbf{Y} - \left(\frac{1}{n}\right) \mathbf{Y}'\mathbf{J}\mathbf{Y}$	$p - 1$	$MSR = \frac{SSR}{p - 1}$
Error	$SSE = \mathbf{Y}'\mathbf{Y} - \mathbf{b}'\mathbf{X}'\mathbf{Y}$	$n - p$	$MSE = \frac{SSE}{n - p}$
Total	$SSTO = \mathbf{Y}'\mathbf{Y} - \left(\frac{1}{n}\right) \mathbf{Y}'\mathbf{J}\mathbf{Y}$	$n - 1$	

F test for regression relation

To test whether there is a regression relation between the response variable Y and the set of X variables $X_1, \dots, X_{p-1}$:

- ① $H_0: \beta_1 = \beta_2 = \dots = \beta_{p-1} = 0$
- ② H_a : not all $\beta_k (k = 1, \dots, p - 1)$ equal zero
- Test statistic:

$$F^* = \frac{MSR}{MSE}.$$

- The decision rule to control the Type I error at α is:
 - ① If $F^* \leq F(1 - \alpha; p - 1, n - p)$, conclude H_0
 - ② If $F^* > F(1 - \alpha; p - 1, n - p)$, conclude H_a

When $p - 1 = 1$ (or $p = 2$), this test reduces to the F test for testing whether or not $\beta_1 = 0$ in simple linear regression.

Coefficient of multiple determination

- The coefficient of multiple determination, denoted by R^2 , is defined as follows:

$$R^2 = \frac{SSR}{SSTO} = 1 - \frac{SSE}{SSTO}$$

- We have $0 \leq R^2 \leq 1$. $R^2 = 0$ when all $b_k = 0 (k = 1, \dots, p - 1)$, and $R^2 = 1$ when all Y observations fall directly on the fitted regression surface, i.e., when $Y_i = \hat{Y}_i$ for all i .
- Adding more X variables to the regression model can only increase R^2 , because SSE is smaller with more X variables added to the model and SSTO is always the same for a given set of responses.

Adjusted coefficient of multiple determination

- Since R^2 can be made larger by including more variables, hence, sometimes the *adjusted coefficient of multiple determination* R_a^2 is used. R_a^2 adjusts R^2 by dividing each sum of squares by its degrees of freedom:

$$R_a^2 = 1 - \frac{SSE}{n - p} \div \frac{SSTO}{n - 1} = 1 - \left(\frac{n - 1}{n - p} \right) \frac{SSE}{SSTO}$$

- Note that a large value of R^2 does not necessarily imply that the fitted model is a useful one.
- The *coefficient of multiple correlation* R is the positive square root of R^2 : $R = \sqrt{R^2}$.

Outline

- 1 Multiple regression model
- 2 Multiple regression model in matrix form
- 3 Analysis of variance results
- 4 Inferences about regression parameters**
- 5 Estimation of mean response and prediction of new observation
- 6 Testing the General Linear Hypothesis
 - Extra sums of squares
 - Testing for a subset of coefficients
- 7 Hidden extrapolation and multicollinearity

Variance-covariance matrix $\text{var}\{\mathbf{b}\}$

- The least squares and maximum likelihood estimators in $\mathbf{b}$ are unbiased:

$$\mathbf{E}\{\mathbf{b}\} = \boldsymbol{\beta}$$

- Variance-covariance matrix σ^2 :

$$\begin{aligned}\text{var}\{\mathbf{b}\} &= \begin{bmatrix} \text{var}\{b_0\} & \text{cov}\{b_0, b_1\} & \dots & \text{cov}\{b_0, b_{p-1}\} \\ \text{cov}\{b_1, b_0\} & \text{var}\{b_1\} & \dots & \text{cov}\{b_1, b_{p-1}\} \\ \vdots & \vdots & & \vdots \\ \text{cov}\{b_{p-1}, b_0\} & \text{cov}\{b_{p-1}, b_1\} & \dots & \text{var}\{b_{p-1}\} \end{bmatrix} \\ &= \sigma^2 * (\mathbf{X}'\mathbf{X})^{-1}\end{aligned}$$

- The estimated variance-covariance matrix:

$$s^2\{\mathbf{b}\} = \text{MSE} * (\mathbf{X}'\mathbf{X})^{-1}$$

Interval estimation of β_k

- For the normal error regression model, we have:

$$\frac{b_k - \beta_k}{s\{b_k\}} \sim t(n - p)$$

- Confidence limit of β_k with $1 - \alpha$ confidence coefficient are:

$$b_k \pm t(1 - \alpha/2; n - p) * s\{b_k\}$$

Tests for individual β_k

- Hypothesis

$$H_0 : \beta_k = 0$$

$$H_a : \beta_k \neq 0$$

- Test statistic $t^* = \frac{b_k}{s\{b_k\}}$.
- If $|t^*| \leq t(1 - \alpha/2; n - p)$, conclude H_0 . Otherwise, conclude H_a .

- The Bonferroni joint confidence intervals can be used to estimate several regression coefficients simultaneously.
- If g parameters are to be estimated jointly (where $g \leq p$), the confidence limits with confidence coefficient $1 - \alpha$ are:

$$b_k \pm B * s\{b_k\},$$

where $B = t(1 - \alpha/2g; n - p)$.

Outline

- 1 Multiple regression model
- 2 Multiple regression model in matrix form
- 3 Analysis of variance results
- 4 Inferences about regression parameters
- 5 Estimation of mean response and prediction of new observation**
- 6 Testing the General Linear Hypothesis
 - Extra sums of squares
 - Testing for a subset of coefficients
- 7 Hidden extrapolation and multicollinearity

Interval estimation of $E\{Y_h\}$

Given the values $X_{h1}, \dots, X_{h,p-1}$, $X_h = [1 \ X_{h1} \ \dots \ X_{h,p-1}]'$, the mean response is $E\{Y_h\} = X_h'\beta$. Hence, the estimated mean response corresponding to X_h is

$$\hat{Y}_h = X_h' b$$

- Mean: $E\{\hat{Y}_h\} = X_h'\beta = E\{Y_h\}$ (unbiased estimator)
- Variance: $\text{var}\{\hat{Y}_h\} = X_h'\text{var}\{b\}X_h = \sigma^2 X_h'(X'X)^{-1}X_h$
Hence, estimated variance is:
 $s^2\{\hat{Y}_h\} = X_h's^2\{b\}X_h = \text{MSE} * X_h'(X'X)^{-1}X_h.$
- CI for $E\{\hat{Y}_h\}$ with $1 - \alpha$ confidence are:

$$\hat{Y}_h \pm t(1 - \alpha/2; n - p) * s\{\hat{Y}_h\}.$$

Prediction of new observation $Y_{h(new)}$

The $1 - \alpha$ prediction limits for a new observation $Y_{h(new)}$ corresponding to X_h are:

$$\hat{Y}_h \pm t(1 - \alpha/2; n - p) * s\{pred\},$$

where $s^2\{pred\} = MSE + s^2\{\hat{Y}_h\} = MSE * (1 + X_h'(X'X)^{-1}X_h)$.

Simultaneous CI for several mean responses

- Working-Hotelling procedure: $1 - \alpha$ confidence band for the regression surface at several X_h are:

$$\hat{Y}_h \pm W * s\{\hat{Y}_h\},$$

where $W^2 = pF(1 - \alpha; p, n - p)$.

- Bonferroni procedure: when estimating g interval

$$\hat{Y}_h \pm B * s\{\hat{Y}_h\},$$

where $B = t(1 - \alpha/2g; n - p)$.

- Which procedure yields a smaller W or B will be used.

Outline

- 1 Multiple regression model
- 2 Multiple regression model in matrix form
- 3 Analysis of variance results
- 4 Inferences about regression parameters
- 5 Estimation of mean response and prediction of new observation
- 6 Testing the General Linear Hypothesis
 - Extra sums of squares
 - Testing for a subset of coefficients
- 7 Hidden extrapolation and multicollinearity

Extra sums of squares

- An *extra sum of squares* measures the marginal reduction in the error sum of squares when one or several predictor variables are added to the regression model, given that other predictor variables are already in the model.
- Example: a study of the relation of amount of body fat (Y) to several possible predictor variables, based on a sample of 20 healthy females 25-34 years old. Possible predictor variables are
 - triceps skinfold thickness (X_1),
 - thigh circumference (X_2), and
 - midarm circumference (X_3)

Subject	Triceps Skinfold Thickness	Thigh Circumference	Midarm Circumference	Body Fat
i	X_{i1}	X_{i2}	X_{i3}	Y_i
1	19.5	43.1	29.1	11.9
2	24.7	49.8	28.2	22.8
3	30.7	51.9	37.0	18.7
...	...	...	...	...
18	30.2	58.6	24.6	25.4
19	22.7	48.2	27.1	14.8
20	25.2	51.0	27.5	21.1

Example - some regression models

Several model to regress body fat (Y):

① Regression of Y on X_1 :

- Model $\hat{Y} = -1.496 + 0.8572X_1$
- Regression sum of squares $SSR(X_1) = 352.27$; error sum of squares $SSE(X_1) = 143.12$

② Regression of Y on X_2 :

- Model $\hat{Y} = -23.634 + 0.8565X_2$
- Regression sum of squares $SSR(X_2) = 381.97$; error sum of squares $SSE(X_2) = 113.42$

③ Regression of Y on X_1, X_2 :

- Model $\hat{Y} = -19.174 + 0.2224X_1 + 0.6594X_2$
- Regression sum of squares $SSR(X_1, X_2) = 385.44$; error sum of squares $SSE(X_1, X_2) = 109.95$

④ Regression of Y on X_1, X_2, X_3 :

- Model $\hat{Y} = 117.08 + 4.334X_1 - 2.857X_2 - 2.186X_3$
- Regression sum of squares $SSR(X_1, X_2, X_3) = 396.98$; error sum of squares $SSE(X_1, X_2, X_3) = 98.41$

- Extra sum of squares:

$$\begin{aligned} SSR(X_2|X_1) &= SSE(X_1) - SSE(X_1, X_2) = 143.12 - 109.95 = 33.17 \\ &= SSR(X_1, X_2) - SSR(X_1) = 385.44 - 352.27 = 33.17 \end{aligned}$$

This reduction in the error sum of squares is the result of adding X_2 to the regression model when X_1 is already included in the model.

- Similarly, $SSR(X_3|X_1, X_2) = SSE(X_1, X_2) - SSE(X_1, X_2, X_3) = 109.95 - 98.41 = 11.54$
- We already know $SSTO = \sum_i (Y_i - \bar{Y})^2 = SSR + SSE$, hence, the reduction in SSE implies an identical increase in SSR.

Decomposition of SSR into Extra Sums of Squares

$$SSR(X_1, X_2) = SSR(X_1) + SSR(X_2|X_1)$$

$$= SSR(X_2) + SSR(X_1|X_2)$$

$$SSR(X_1, X_2, X_3) = SSR(X_1) + SSR(X_2|X_1) + SSR(X_3|X_1, X_2)$$

$$= SSR(X_2) + SSR(X_3|X_2) + SSR(X_1|X_2, X_3)$$

$$= SSR(X_1) + SSR(X_2, X_3|X_1)$$

The columns ($\mathcal{B}$) is divided into two subsets: $\mathcal{B}_1 : n \times (p - r)$, $\mathcal{B}_2 : n \times r$
We want to find the contribution of the terms in $\mathcal{B}_2$ to the regression
 $H_0 : \mathcal{B}_2 = 0$.

- Full model: includes all the columns, $\mathcal{B}$
- Reduced model: the model with the subset $\mathcal{B}_1$
- Extra sum of squares due to $\mathcal{B}_2$:

$$SSR(\mathcal{B}_2|\mathcal{B}_1) = SSR(\mathcal{B}) - SSR(\mathcal{B}_1) = SSE(\mathcal{B}_1) - SSE(\mathcal{B})$$

- Test for $H_0 : \mathcal{B}_2 = 0$:

$$F^* = \frac{SSR(\mathcal{B}_2|\mathcal{B}_1)/r}{MSE(\mathcal{B})}$$

Summary of tests for the coefficients

- Test for all coefficients: $H_0 : \beta_0 = \beta_1 = \dots = \beta_{p-1} = 0$
- Test for a subset of coefficients
- Test for a single coefficient $H_0 : \beta_k = 0$.

Outline

- 1 Multiple regression model
- 2 Multiple regression model in matrix form
- 3 Analysis of variance results
- 4 Inferences about regression parameters
- 5 Estimation of mean response and prediction of new observation
- 6 Testing the General Linear Hypothesis
 - Extra sums of squares
 - Testing for a subset of coefficients
- 7 Hidden extrapolation and multicollinearity

Hidden extrapolate in multiple regression

- One must be careful about extrapolating beyond the region containing the original observations. The model fits well in the region of the original data will perform poorly outside that region.

- If

$$h_{00} = x_0^T (X^T X)^{-1} x_0 \geq h_{\max},$$

where $h_{\max}$ is the largest value of h_{ii} , then x_0 is extrapolation points. Although x_0 may lie within the ranges of the regressors.

- See more details in Section 3.9, MPV book.

Multicollinearity

- A serious problem that may dramatically impact the usefulness of a regression model is **multicollinearity**, or **near-linear dependence** among the regression variables.
- The main diagonal elements of the inverse of the $(X^T X)^{-1}$ matrix in correlation form (see section 3.10, MPV) are often called **variance inflation factors** (VIFs).
- Let's R_j^2 be the coefficient of multiple determination obtained from regressing x_j on the other regressor variables

$$VIF_j = \frac{1}{1 - R_j^2}.$$

- VIFs larger than 10 imply serious problems with multicollinearity.

A quiz about of Multiple Regression.

Summary

- Multiple regression model
- Analysis of variance
- Inferences about regression parameters
- Estimation of Mean Response and Prediction of New Observation
- Testing the General Linear Hypothesis
- An example of Multiple Regression

THANK YOU FOR YOUR ATTENTION!